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Transverse vertical dispersion in groundwater and the capillary fringe

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Abstract

Transverse dispersion is the most relevant process in mass transfer of contaminants across the capillary fringe (both directions), dilution of contaminants, and mixing of electron acceptors and electron donors in biodegrading groundwater plumes. This paper gives an overview on literature values of transverse vertical dispersivities α_{tv} measured at different flow velocities and compares them to results from well-controlled laboratory-tank experiments on mass transfer of trichloroethene (TCE) across the capillary fringe. The measured values of transverse vertical dispersion in the capillary fringe region were larger than in fully saturated media, which is credited to enhanced tortuosity of the flow paths due to entrapped air within the capillary fringe. In all cases, the values observed for α_{tv} were $\ll 1$ mm. The new measurements and the literature values indicate that α_{tv} apparently declines with increasing flow velocity. The latter is attributed to incomplete diffusive mixing at the pore scale (pore throats). A simple conceptual model, based on the mean square displacement and the pore size accounting for only partial diffusive mixing at increasing flow velocities, shows very good agreement with measured and published data.
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Keywords: Groundwater risk assessment; Organic pollutants; Capillary fringe; Mixing; Transverse dispersivity

1. Introduction

1.1. Transverse dispersion/mixing in porous media

The quantification of mixing rates in porous media is a prerequisite for groundwater risk assessment (e.g. transport of contaminants into the groundwater across the capillary fringe),

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for remediation (e.g. dissolution kinetics of NAPL; Ahmadi et al., 2001), and natural attenuation of contaminant plumes (e.g. dilution and biodegradation, the rates of which may depend on the transfer rates of electron acceptors or donors into the plume) (Kitanidis, 1994; Cirpka et al., 1999). Furthermore, transverse mixing reduces plume "fingering" caused by longitudinal dispersion in heterogeneous porous media (Dagan, 1991; Cirpka and Kitanidis, 2000; Dentz et al., 2000). Transverse dispersion is in general important for mass transfer processes in packed beds (Guedes de Carvalho and Delgado, 1999). Transverse mixing consists of two components: molecular diffusion of the solute and the tortuous flow paths of water in the porous medium. The objective of this paper is to summarise data on transverse dispersion from different laboratory and field studies (see Fig. 1 and Table 1 for an overview) and to compare these with results from a well-controlled tank experiment on transport of TCE across the capillary fringe at increasing flow velocities. In the following, we provide a brief overview on the experiments used to determine transverse dispersivities from the literature.

1.2. Transverse dispersivities from transport of volatile compounds across the capillary fringe

In the literature, a couple of experiments can be found dealing with the volatilisation of volatile compounds from groundwater. All of them were conducted with trichloroethene (TCE) in laboratory tanks. Swallow and Gschwend (1983) report experiments in sand at a groundwater velocity of 0.84 m day $^{-1}$. They obtained a transverse vertical dispersivity of 3.3 cm. McCarthy and Johnson (1993) measured the volatilisation in Ottawa sand (mean grain diameter of 1 mm) at a groundwater velocity of 0.1 m day $^{-1}$. In contrast to Swallow

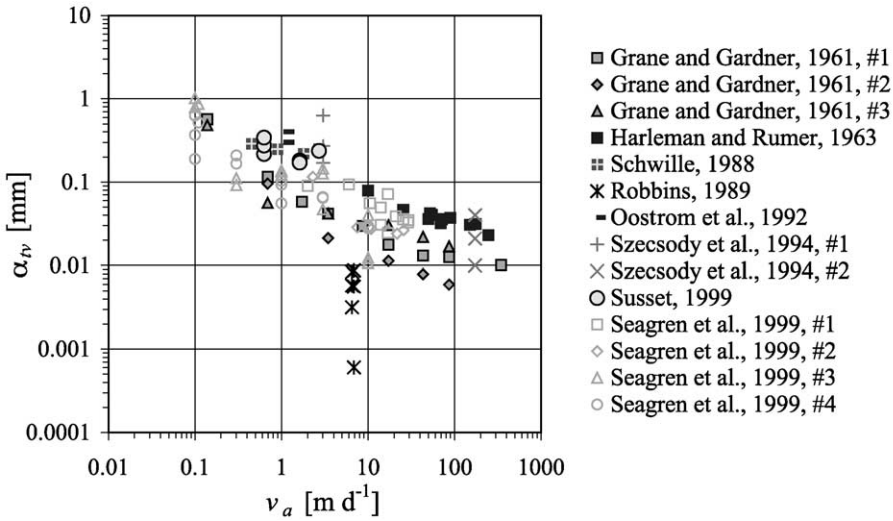

Fig. 1. Transverse vertical dispersivities α_{tv} vs. flow velocity. All data were obtained in fully saturated media, except data from Oostrom et al. (1992) and Susset (1998), which result from transport of TCE across the capillary fringe. Parameters used for calculation of α_{tv} are compiled in Table 1.

Table 1
Parameters used for the calculation of transverse dispersivities shown in Figs. 1 and 5

Reference		v_a (m day^{-1})	d (cm)	D_{aq} n (cm^2 s^{-1})	α_{tv} (mm)
Grane and Gardner (1961)	#1	0.14–345.6	0.025	3.4×10^{-7}	0.010–0.57
	#2	0.69–86.4	0.0074	3.2×10^{-7}	0.006–0.096
	#3	0.14–86.4	0.15	3.1×10^{-7}	0.017–0.48
Harleman and Rumer (1963)		10.02–245.4	0.096	6.5×10^{-7}	0.023–0.079
Robbins (1989)		6.53–6.86	0.048	5.8×10^{-6}	0–0.0088
Oostrom et al. (1992)		1.08	0.05	4.0×10^{-8}	0.3–0.4
Schwille (1988), cited in Johnson and Pankow (1992)		0.45–2.7	0.02–0.1	2.7×10^{-6}	0.22–0.29
Szecsody et al. (1994)	#1	3.02	0.015	8.1×10^{-7}	0.17–0.63
	#2	172.8	0.06	8.2×10^{-7}	0.01–0.04
Susset (1998)		0.63–2.72	0.2–0.4	3.4×10^{-6}	0.17–0.34
Seagren et al. (1999)	#1	2–29.2	0.2	2.9×10^{-5}	0.024–0.094
	#2	2.3–25.7		2.9×10^{-5}	0.024–0.12
	#3	0.1–10.0		2.9×10^{-5}	0.011–1.0
	#4	0.1–10.1		3.1×10^{-5}	0.028–0.63

If not reported, α_{tv} was obtained from the dispersion coefficient based on Eq. (4) ($\alpha_{tv}=(D_{tv} - D_{aq}n)/v_a$).

and Gschwend (1983), they found that at low groundwater velocities (e.g. 0.1 m day^{-1}), diffusion rather than mechanical dispersion is the dominant transport process. Susset (1998) studied the fluxes of TCE from groundwater to the unsaturated zone for a fine gravel at groundwater flow velocities in the range between 0.63 and 2.72 m day^{-1} obtaining a transverse dispersivity value of 0.226 mm.

Only three recent experiments were found on the mass transfer of a compound from the unsaturated zone to groundwater. One study was performed by Caron et al. (1998) in a laboratory tank with carbon dioxide (CO_2) in a dune sand at a groundwater flow velocity of about 0.25 m day^{-1}. The authors did not calculate transverse dispersivities. Another experiment was done by Jellali (2000) in a fine to medium sand for mass transfer of TCE from soil–gas to groundwater at a groundwater flow velocity of 0.1 m day^{-1}. Jellali (2000) found a vertical transverse dispersivity of 0.43 mm. Rivett (1995) concluded that mass transfer of trichloroethene and tetrachloroethene from soil–gas to the groundwater at the Borden field site at a flow velocity of 0.4 m day^{-1} was controlled by diffusion (i.e. $\alpha_{tv} = 0$).

1.3. Transverse dispersivities from pool dissolution experiments

Transverse dispersivities can also be calculated from data on dissolution rates of NAPL pools measured in laboratory tank experiments (Table 1). Schwille (1988) used trichloroethene (TCE) in medium to coarse sand and flow velocities ranging from 0.45 to 2.7 m day^{-1}; Pearce et al. (1994) used TCE and trichloroethane (TCA) in medium-grained silica sand for groundwater flow velocities in the range of 0.08–0.09 m day^{-1}; Oostrom et al. (1999) investigated TCE pools in a coarse sand at a flow velocity of 0.44 m day^{-1} and Seagren et al. (1999) worked with toluene and a mixture of toluene and dodecane in a porous medium of soda-lime glass beads at velocities from 0.1 to 29.2 m day^{-1}. They found α_{tv} in the order of less than 0.1 mm. This is confirmed by experimental observations from Loyek (1998) and Grathwohl et al. (2000) on the release of organic pollutants from

coal tar pools. At the field-scale, a dissolution experiment was performed by Schäfer and Therrien (1995) in coarse sand at a flow velocity of 0.48 m day^{-1}. Note that the boundary condition for dissolution of pools is the same as for diffusion of contaminants from the unsaturated zone into groundwater (but in general, lower values for the transverse dispersivity are observed) and for mass transfer from buried flat wall in a porous medium in general (Delgado and Guedes de Carvalho, 2001).

1.4. Transverse dispersivities from tracer experiments (laboratory and field)

Tracer experiments for the determination of transverse dispersivities in water-saturated porous media were conducted by many researchers. In the following, some typical examples for experiments in laboratory tanks are given (see Table 1 for a summary of the results). Grane and Gardner (1961) conducted dispersion experiments with three types of glass beads (mean diameter of 0.0074, 0.025 and 1.5 mm) at groundwater velocities between 0.14 and 345.6 m day^{-1}. Harleman and Rumer (1963) measured the dispersion of sodium chloride for laminar water flow in beds of plastic spheres with a mean diameter of 0.96 mm and a porosity of 0.36 at water velocities ranging from 10 to 245 m day^{-1}. Oostrom et al. (1992) studied the dispersion of sodium iodide in Ottawa quartz sand at a groundwater velocity of 1.08 m day^{-1}. Robbins (1989) measured transverse dispersion of bromide in glass beads at water velocities between 6.53 and 6.86 m day^{-1} and Szecsody et al. (1994) studied the dispersive behaviour of calcium chloride at a flow velocity of 3.02 m day^{-1} in fine sand and at a flow velocity of 172.8 m day^{-1} in medium sand. Guedes de Carvalho and Delgado (2000) observed in packed beds of coarse sand a nonlinear relationship between the transverse dispersion coefficient and the flow velocities at Peclet numbers above 70.

Field tracer tests were performed for example by Moltyaner and Killey (1988) with iodide in fluvial sands at a groundwater velocity of 1.2 m day^{-1}; Sudicky et al. (1983) performed a tracer experiment with chloride in a sandy aquifer (horizontally bedded fine- to medium-grained sands with a porosity of 0.38) at the Borden site. They stated that the overall transverse vertical dispersion on the site was very weak and in the order of the aqueous diffusivity. Results from numerical modelling of the Borden tracer tests data indicate that the transverse macrodispersivity for the site is 2.2 mm (Rajaram and Gelhar, 1991). Field studies on the Cap Code site report a transverse macrodispersivity for this site of 1.5 mm (Garabedien et al., 1988). Transport simulations accounting for the heterogeneity of the medium yield much lower values of 0.5 mm for the pore scale dispersivity (Hantush and Mariño, 1998) and 0.44 mm (Fiori and Dagan, 1999). Such small values (0.5 mm) were also found in a sandy aquifer at the Vejen site in Denmark (Jensen et al., 1993).

2. Experimental methods

2.1. Mass transfer across the capillary fringe

Bench-scale tank experiments were performed in order to elucidate relationships between TCE mass transfer rates across the capillary fringe and the flow velocity in a coarse sand. The

experiment was conducted in a glass tank, which was 150 cm long, 27.5 cm wide and 58 cm high as shown in Fig. 2. Vertical sheet piles of stainless steel were installed 15 cm from each end of the tank reaching from just below the water table (21 cm measured from the bottom of the tank) up to the top of the tank. The region from the base of the tank to a height of 21 cm was screened with a stainless steel mesh. Hence, water could flow through the entire saturated zone, while a shortcut of the headspace in the inlet/outlet chambers via the unsaturated zone was prevented by the sheet piles. Water was pumped into the inlet reservoir with a peristaltic pump; in the outflow reservoir, a constant water table was established by allowing the water flowing out of a steel tube ending at a gauged depth outside the tank.

The mean inlet and outlet concentrations were obtained by sampling water at T-fittings (each closed with a septum) with a gas-tight syringe. Gas samples from the unsaturated zone of the tank were taken from multilevel stainless steel tubing installed in regular distances within the tank (see Fig. 2) and ending in depths of 10, 20 and 30 cm below the top of the tank.

The tank (excluding inlet and outlet chambers) was filled with a medium to coarse sand from a gravel pit in the Upper River Rhine Valley, Germany. The sand consists mainly of silicates (quartz and feldspars) with a minor fraction of carbonates. The carbonate content of the sand was 2.8% and the fraction of organic carbon fraction (f_{oc}) was 0.0001 (determined after acid treatment by dry combustion under oxygen and subsequent carbon dioxide analysis). The grain size was between 0.3 and 2 mm with an average of 0.7 mm (50% value taken from the standard gravimetric sieve analysis). The sand was carefully filled into the tank in layers of 0.5 cm. During the whole filling procedure, the water table was adjusted to just above the "working surface" in order to minimise the entrapment of air bubbles in the pores. From a tracer experiment with sodium naphthionate, a flow-effective porosity of 0.356 was calculated for the sand. The calculation of the porosity n from the mass M of a sand

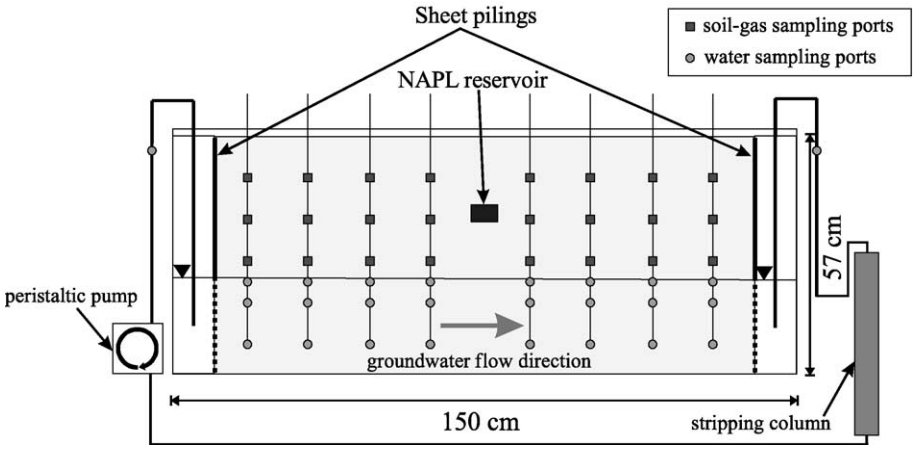

Fig. 2. Setup of the tank experiment for mass transfer of TCE from soil–gas to groundwater.

portion and its volume V ($n = 1 - M/(\rho_g V)$), using a solid density ρ_g of 2.65 g cm^{-3} yielded a value for the overall porosity of 0.354, which is in excellent agreement with the results from the tracer test.

Before the experiment was started, the top of the glass tank was sealed with a stainless steel plate. For conditioning the sand and removing suspended matter, deionized water was flushed through the tank for 3 weeks prior to the experiment. A 0.5 m tall air-stripping tower was installed at the outlet of the tank allowing the cleanup of water leaving the tank and circulating it during the experiments from outlet to inlet.

During the run of the experiment, water was flushed through the tank at different average flow velocities (v_a) ranging from 1.53 to 11.33 m day^{-1}. It should be noted that the variation of the flow velocities resulted in changes of the height of the water table with the likely consequence of the entrapment of air bubbles in the capillary fringe region. Since these entrapped air bubbles are isolated from each other, gas diffusion does not contribute significantly to the overall diffusive flux (diffusion in series: the effective diffusion coefficient is proportional to $1/(\delta_g/(HD_g) + \delta_w/D_w)$; D_g, D_w, δ_g and δ_w denote diffusion coefficients and the distances traveled in gas and water, respectively; H is Henry's law constant; since δ_g certainly is small—only few gas bubbles were observed—and D_g is relatively large, aqueous diffusion will still dominate overall mass transfer). The entrapped air bubbles, however, will likely result in increased tortuosity of flow. In addition, they act as "mixing chambers" because of the fast diffusion in the gas phase. Both effects can lead to an overall increase of the dispersivity in the capillary fringe region. The temperature of the water in the tank was monitored at the outlet and was within 24.4 ± 0.5 °C. Trichloroethene (TCE) was provided to the unsaturated zone over glass tubes ending in a cylinder with an open upper side (cross-sectional area of 10.2 cm^2) in the middle of the tank at a depth of 20 cm. The cylinder, with a volume of about 30 cm^3, was acting as a TCE reservoir and had a level gauge installed at the glass wall indicating the degree of filling during the experiment. In addition, small amounts of TCE (< 10 μl) were carefully injected in the upper part of the unsaturated zone near the inlet and outlet reservoirs with a syringe (injected TCE volumes were low enough to avoid contamination of underlying groundwater with DNAPL). This procedure resulted in a homogeneous concentration of TCE in the unsaturated zone during the whole duration of the experiment as confirmed by vapour-phase measurements.

2.2. Sampling and analyses

Gas samples from the multilevel stainless steel tubing were taken from time to time with a 100 μl gas-tight syringe at depths of 10, 20 and 30 cm in the unsaturated zone. The gas-tight syringe (with a long steel needle reaching into the steel tubing) and a bigger syringe (10 ml) were attached to the capillary through a T-connection and Teflon tubing. To purge the sampling tube, a volume of two times the tubing volume was displaced with the bigger syringe. Then, a gas sample of 50 μl was taken with the gas-tight syringe and analysed directly with GC-FID. The measured TCE concentrations in the gaseous phase were always above 80% and close to the vapour saturation. The upper boundary condition for the evaluation of the mass transfer across the capillary fringe, therefore, was saturation vapour concentration throughout the entire unsaturated zone.

The TCE concentration in the exhaust air of the stripping tower receiving the contaminated "groundwater" from the tank was measured on-line with a portable gas-photometer every 45 s. The gas-phase concentration resulting from the stripping tower is directly related to the aqueous concentration at the outlet chamber of the tank (accounting for the water and gas flow rates in the stripping tower and the stripping rate). The measured on-line curve was calibrated with water samples taken at the outlet of the tank from time to time. This procedure allowed monitoring continuously the concentration at the outlet of the tank. At steady state conditions, the total TCE flux leaving the tank is equal to the mass flux through the capillary fringe:

$$F_{\text{tank}} = C_{\text{gw,out}} Q = C_{\text{gw,out}} A v_{\text{a}} n \tag{1}$$

where F_{tank} (mg s^{-1}) is the contaminant flux leaving the tank, $C_{\text{gw,out}}$ (mg l^{-1}) is the measured mean aqueous concentration at the outlet of the tank, Q (l s^{-1}) is the water flow rate through the tank, and A is the cross-sectional area of the saturated zone (cm^2) (equals the height of water-saturated zone h_{gw} (cm) times the width of tank B (cm)). For analysis and evaluation of the experiment, h_{gw} was estimated as:

$$h_{\text{gw}} = \frac{h_{\text{gw}}^{\text{in}} + h_{\text{gw}}^{\text{out}}}{2} \tag{2}$$

where $h_{\text{gw}}^{\text{in}}$(cm) is the height of the water table in the inlet chamber and $h_{\text{gw}}^{\text{out}}$ (cm) is the height of the water table in the outlet chamber of the tank ($h_{\text{gw}}^{\text{in}} - h_{\text{gw}}^{\text{out}}$ was always less than 1% of the water-saturated zone h_{gw}).

3. Determination of transverse dispersivities

The mass transfer F (g day^{-1}) from a flat wall into a packed bed (e.g. dissolution of an NAPL pool or diffusion across the capillary fringe) at the second boundary condition is given by an analytical solution of Fick's second law, the so-called surface renewal model (e.g. Johnson and Pankow, 1992; Guedes de Carvalho and Delgado, 1999; Delgado and Guedes de Carvalho, 2001):

$$F = C_{\text{w}} n \sqrt{\frac{4}{\pi} D_{\text{tv}} \frac{v_{\text{a}}}{L}} BL \tag{3}$$

$$\text{with } D_{\text{tv}} = \alpha_{\text{tv}} v_{\text{a}} + D_{\text{aq}} n \tag{4}$$

where C_{w} is the constant concentration of the contaminant at the interface (air/water or NAPL/water), D_{tv} is the transverse vertical hydrodynamic dispersion coefficient, L and B are the length and width of the interfacial area ($L \times B$ = contaminated area of soil–gas or pool surface). D_{aq} and n denote the diffusion coefficient in water and porosity. $D_{\text{aq}}\, n$ represents the pore diffusion coefficient assuming that the tortuosity factor is approximately proportional to n^{-1} (Grathwohl, 1998). Note that similar expressions were derived for the estimation of Sherwood number as a function of flow velocities (Guedes

de Carvalho and Delgado, 1999). By rearranging Eqs. (3) and (4), the transverse vertical dispersivity α_{tv} can be obtained from the measured steady state flux:

$$\alpha_{tv} = \frac{\left(\frac{F}{2C_w Bn}\right)^2 \frac{\pi}{Lv_a} - D_{aq}n}{v_a} \tag{5}$$

With this evaluation, it is assumed that (1) the interface between the saturated and the unsaturated zone is flat, (2) the flow is parallel to the interface, and (3) the flow velocity as well as α_{tv} are uniform within the mass transfer zone. While assumptions 1 and 2 are certainly valid due to the experimental setup (coarse homogeneous sand, low interfacial tension because of the high TCE concentrations), the flow velocity as well as α_{tv} probably change within the effective mass transfer zone (which is approximately 3 cm thick). Therefore, only lumped values for α_{tv} are determined. Since the flow velocity in the capillary fringe is lower than the average flow velocity in the tank, α_{tv} tends to be underestimated at a given measured flux (F in Eq. (5)).

4. Results and discussion

4.1. Transverse dispersivity from tank experiments on TCE transport across the capillary fringe

The TCE concentrations obtained from on-line measurements are shown in Fig. 3. Changes in the water flow velocity resulted in fluctuations of the water table and therefore, varying TCE concentrations with time at the outlet of the tank. Steady state conditions (necessary for the determination of α_{tv}) were obtained after several pore volumes of water in the saturated zone (until TCE concentrations at the tank outlet were stable again). For the calculation of the TCE fluxes to the groundwater, only concentrations from the steady state part of the on-line curve were taken (see Fig. 3).

The diffusive–dispersive TCE fluxes at various flow rates were calculated (Eq. (1)) using an aqueous diffusion coefficient of 8.38×10^{-5} m^2 day^{-1} (calculated from Worch, 1993) and an aqueous solubility of TCE of 1322.5 mg l^{-1} (Verschueren, 1996) at the interface. The parameters used for calculation are listed in Table 2.

Fig. 4 compares the measured TCE fluxes to fluxes expected from Eq. (3) (solid line in Fig. 4) at increasing flow velocities. A linear relationship between flow velocities and TCE fluxes was only obtained at flow velocities <4 m day^{-1}, corresponding to a value of the transverse vertical dispersivity of 0.63 mm. At higher flow velocities, the measured TCE fluxes were significantly smaller than that expected from Eq. (3) indicating an apparently decreasing dispersivity with increasing flow velocity. This trend is also clearly visible from the literature data shown in Fig. 1, which is in contrast to earlier work. According to de Josselin de Jong (1958), the transverse dispersivity α_{tv} can be correlated to the characteristic length l_{ch} of the porous medium:

$$\alpha_{tv} = \frac{3}{16} l_{ch} \tag{6}$$

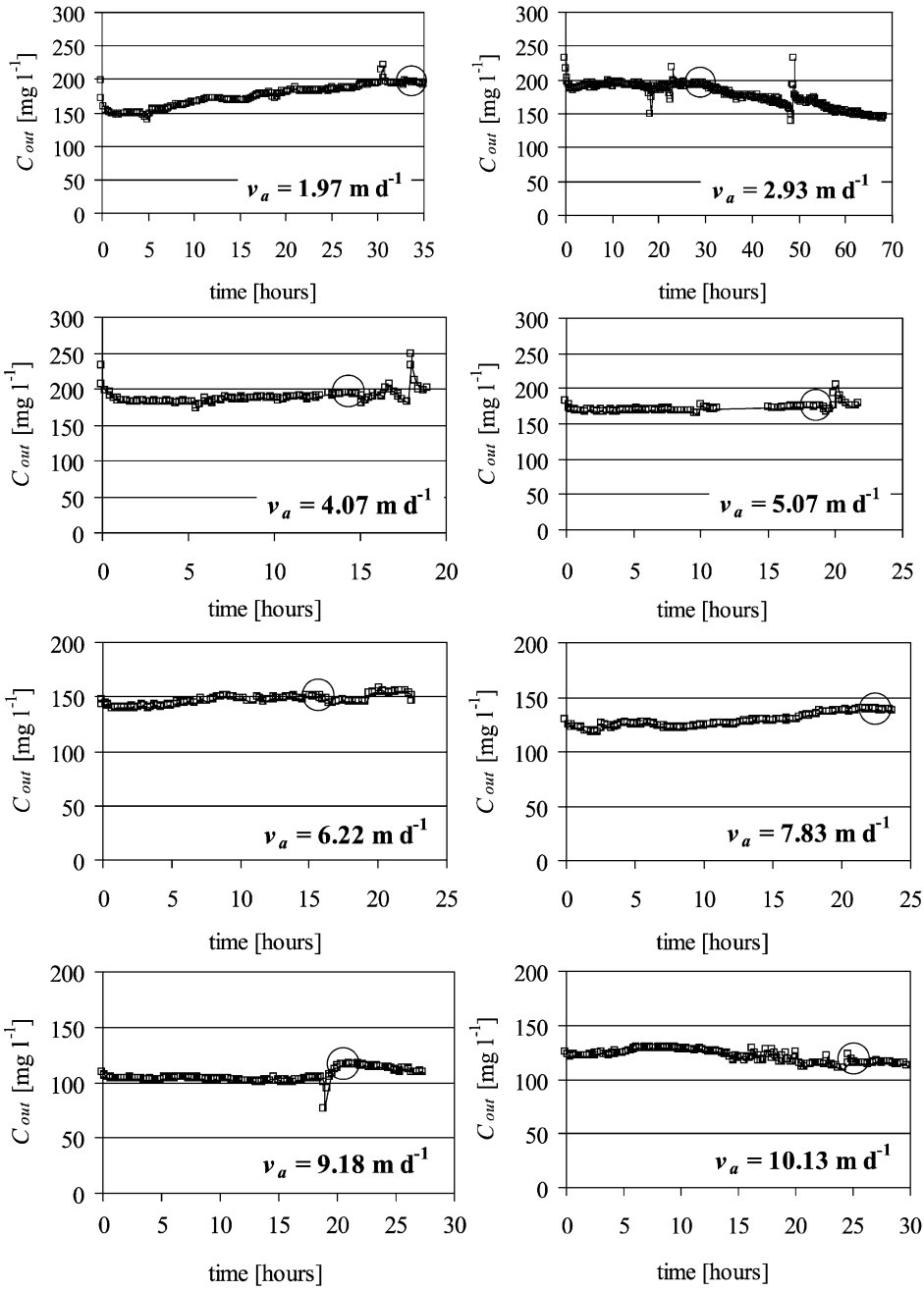

Fig. 3. On-line measurements of TCE concentrations in water in the outlet chamber of the tank experiment on mass transfer across the capillary fringe; large circles denote time periods where the steady-state concentrations were obtained for the determination of α_{tv} (results in Table 2).

Table 2
Steady state TCE concentrations at the outlet of the tank and calculated fluxes from the unsaturated to the saturated zone for various groundwater velocities used for the determination of α_{tv}

v_a (m day^{-1})	Q (m^3 day^{-1})	$C_{gw,out}$ (mg l^{-1})	F_{tank} (g day^{-1})	α_{tv} (mm)
11.33	0.258	116	29.98	0.251
10.13	0.229	118	27.02	0.255
7.11	0.163	116	18.94	0.253
6.06	0.139	138	19.24	0.360
4.82	0.109	150	16.36	0.411
3.93	0.087	175	15.28	0.541
3.15	0.069	195	13.43	0.649
2.27	0.049	195	9.56	0.631
1.53	0.033	195	6.37	0.610

Eq. (6) was derived from a statistical model in which the porous medium is represented by interconnected channels of equal length, which are uniformly distributed in all directions. Measuring dispersion in column experiments, de Josselin de Jong (1958) found that l_{ch} is a function of the grain size distribution and in the order of the mean grain diameter d. Saffman (1959) came to the same result, but he expresses doubts on the validity of some of the assumptions made in the model. Meanwhile, however, Eq. (6) is generally accepted, e.g. in the recent literature on chromatography (Baumeister et al., 1995). One underlying assumption of this model is complete mixing of water in the pore throats due to molecular diffusion (de Josselin de Jong, 1958; Saffman, 1960). We hypothesize that the apparent decrease of α_{tv} with increasing flow velocities (Figs. 1

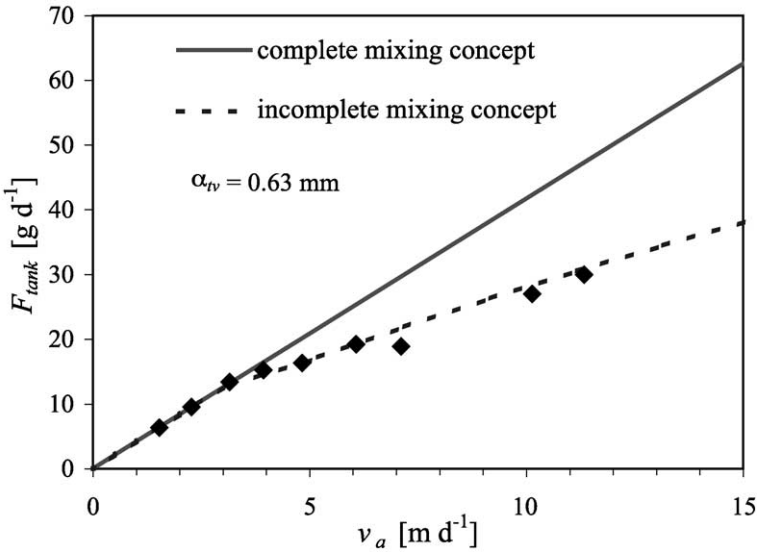

Fig. 4. TCE flux across the capillary fringe at increasing flow velocities. The α_{tv} was fitted to 0.63 mm at flow velocities below 4 m day^{-1}. Solid line: linear increase of the dispersion coefficient with v_a; dashed line: "incomplete mixing" (Eq. (11)) with $d/a = 8$.

and 4) may be due to incomplete equilibration of adjacent streamtubes because of too short contact times in pore throats. In a first approximation, it is assumed that mixing in laminar flow pores may be more or less complete if the effective diffusion distance z (here represented by the mean square displacement) has grown to a comparable size to the radius of a pore a:

$$z = \sqrt{2D_{aq}\frac{l_{ch}}{v}} \tag{7}$$

l_{ch}/v represents the contact time (or mean residence time) and v and l_{ch} denote the velocity in the pore and the length of the pore channel, respectively. As a consequence, above a critical flow velocity v^{crit} at which z becomes less than the radius of the pore a, mixing in the pore throat becomes incomplete. Here, v^{crit} is defined as:

$$v^{crit} = \frac{2D_{aq}l_{ch}}{a^2} \tag{8}$$

For $v > v^{crit}$, a simple correction factor for α_{tv} can be introduced accounting for partial mixing in the pore throat (see Appendix A for the derivation of f_{corr}):

$$f_{corr} = \frac{z}{a} = \frac{\sqrt{D_{aq}\frac{l_{ch}}{v}\frac{4}{\pi}}}{a} \tag{9}$$

Based on the correlation of $\alpha_{tv} = 3/16\ l_{ch}$ from de Josselin de Jong (1958), a corrected transverse dispersivity $\alpha_{tv}{}^{corr}$ for $v > v^{crit}$ can be obtained (for a more detailed derivation, see Appendix A):

$$\alpha_{tv}^{corr} = \frac{3}{16}l_{ch}\,f_{corr} = \frac{3}{16}\frac{l_{ch}}{a}\sqrt{D_{aq}\frac{l_{ch}}{v}\frac{4}{\pi}} \tag{10}$$

Note that the local scale parameters l_{ch} and v are unknown, but the general form of Eq. (10) is supposed to hold if macroscopic parameters are used. In the literature, it is generally agreed, that l_{ch} can be replaced by the grain diameter d (Bear, 1972; Saffman, 1960; de Josselin de Jong, 1958); replacing in Eq. (10) furthermore the pore scale velocity v by the average velocity v_a yields:

$$\alpha_{tv}^{corr} = \frac{3}{16}\frac{d}{a}\sqrt{D_{aq}\frac{d}{v_a}\frac{4}{\pi}} \tag{11}$$

Applying this simple correction to the data measured in the glass tank experiment (dashed line in Fig. 4) and fitting of d and a resulted in a very good Pearson's coefficient of correlation of 0.98. From fitting, a ratio of $d/a = 8$ was obtained, which seems to be a reasonable number. In chromatography, a range of d/a from 4 to 20 is reported (Giddings, 1960).

The dispersivity obtained in this tank experiment from the capillary fringe region is significantly higher than in fully saturated media (Fig. 1). Applying Eq. (6) would yield an unreasonable channel length of 3.36 mm, which is 4.8 fold of the grain diameter. Note that

the sand used was very homogeneous with a narrow grain size distribution, leading to a very regular capillary fringe as observed in the preliminary tracer experiments. The high TCE concentration in the vapour phase (saturation) also leads to a decrease of the interfacial tension between air and water and thus, to a reduction of the height of the capillary rise of water (Smith and Gillham, 1999). Vapour-phase diffusion in entrapped air alone cannot account for these slightly enhanced dispersivity values, as discussed above. Therefore, the only reason for the high dispersivity seems to be the entrapped air bubbles, which may cause highly irregular (tortuous) water flow within the capillary fringe as observed by others before (Ronen et al., 1986, 1997).

Fig. 5 compares the velocity dependency of the literature vertical transverse dispersion coefficients in porous media, the de Josselin de Jong relationship (Eq. (6)) and the simple corrections for the incomplete mixing (Eq. (11)) proposed here. Assuming complete mixing in the pore throats (solid line in Fig. 5) overestimates transverse vertical dispersion at high flow velocities, while for the model proposed here (dashed line in Fig. 5), the agreement between data and model prediction is reasonably good (especially for high flow velocities). Values of transverse vertical dispersion obtained from NAPL pool dissolution experiments (Seagren et al., 1999; Grathwohl et al., 2000) are significantly lower than that observed in transport across the capillary fringe. Nevertheless, the trend of the data is the same as for the modelled curve. The experimental data from the literature indicate that in a porous media with irregularly shaped pores, mixing may become incomplete above $Pe \cong 70$. This is in

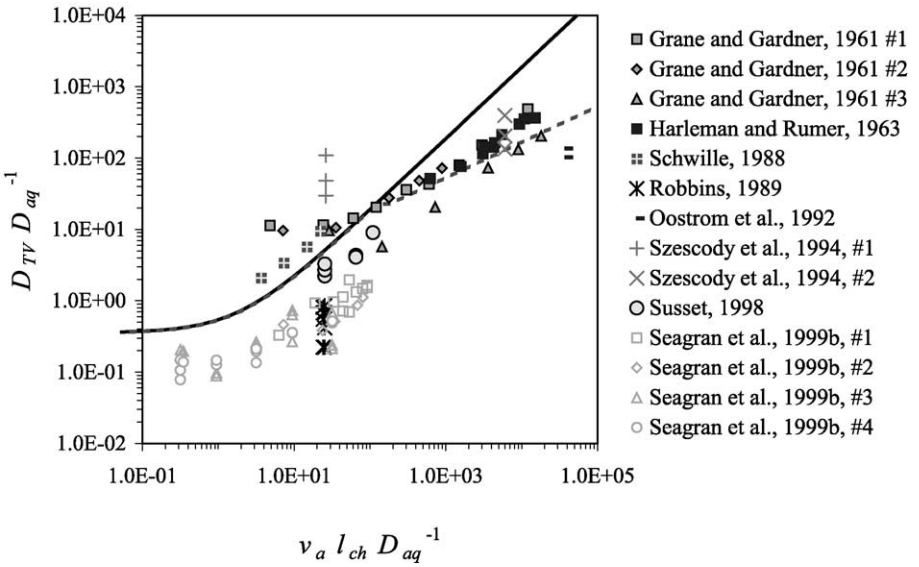

Fig. 5. Relationship between vertical transverse dispersion and flow velocity. The solid line denotes the relationship from de Josselin de Jong (1958) (Eq. (6)). The dashed line accounts for incomplete mixing at the pore scale at Peclet numbers >128 (Eq. (11)). Calculations were done with $d/a = 8$. All data were obtained in fully saturated media, except the data from Oostrom et al. (1992) and Susset (1998), which result from transport across the capillary fringe.

agreement with experimental observations on the velocity dependence of transverse dispersion coefficients by Guedes de Carvalho and Delgado (1999, 2000). Appendix A shows how the cross-sectional concentrations profiles change with dimensionless time (local *Pe* number) in a single pore throat and how the simple approximation developed in Eqs. (7)–(11) can be derived from analytical solutions of Fick's second law.

5. Conclusions

Data from literature and from the tank experiment reported here indicate that values for transverse vertical dispersivity are in general very small, e.g. $\ll 1$ mm. Our measurements of TCE transport across the capillary fringe in a medium- to coarse-grained sand resulted in a α_{tv} value of about 0.63 mm (valid for flow velocities up to 4 m day^{-1}). With this, the contribution of transverse dispersion to the overall mass transfer of volatile compounds across the capillary fringe is significantly higher than from groundwater recharge alone (at a given recharge rate of 0.5 mm day^{-1}, length of the contaminated zone of 10 m and flow velocity of 1 m day^{-1}, an approximately 1.5-cm-thick "layer" of contaminated groundwater forms, whereas the characteristic length for transverse dispersion is approximately 8 cm with $\alpha_{tv} = 0.63$ mm).

The relatively high value obtained for the transverse vertical dispersivity is attributed to irregular flow patterns in the capillary fringe due to entrapped air bubbles. These air bubbles increase the flow tortuosity and they can also act as "mixing chambers" because diffusion in gaseous phase is much faster than in the aqueous phase. Especially with water table fluctuations, various portions of entrapped air may occur in the subsurface environment depending on drying–rewetting cycles and the wetting-history of the capillary fringe.

At flow velocities above about 4 m day^{-1}, an apparent decline in vertical transverse dispersivities was observed in the experiment presented here, as well as in data reported in the literature. This is attributed to incomplete mixing in pore throats, which can be accounted for by the ratio of the mean square displacement to the pore diameter (Eq. (9)). The simple model presented here is of tentative character and certainly more research is necessary to gain a better understanding of the dispersion phenomena in natural porous media especially in the capillary fringe region.

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Appendix A

Assume two parallel streamtubes (2*D*, with a flat interface, Fig. A1), which come into contact in a pore throat of radius *a* for a certain time *t*. Molecular diffusion allows the

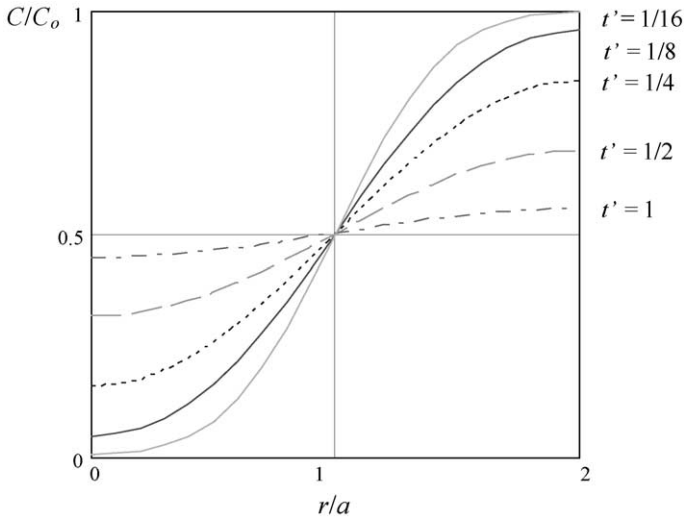

Fig. A1. Change of the concentration profile during diffusion of a solute from one streamtube to another (from right to left). t' denotes dimensionless time.

exchange of solutes from one streamtube to the adjacent one. The concentration profiles which develop are given by an analytical solution of Fick's second law (Crank, 1975):

$$C/C_0 = \frac{1}{2} - \frac{2}{\pi}\sum_{n=0}^{\infty}\left[\frac{(-1)^n}{2n+1}\exp\left(-(2n+1)^2\frac{\pi^2 D_{\mathrm{aq}}t}{4a^2}\right)\cos\left(\frac{2n+1}{2}\pi\frac{r}{a}\right)\right] \tag{A1}$$

r denotes the radial distance to the center of the "pore". As Fig. A1 shows, mixing becomes almost complete as the dimensionless time ($t' = D_{\mathrm{aq}}t/a^2$) approaches 1. The relative mass of solute, which has diffused from one streamtube to the other, is given by:

$$M = \frac{1}{2} - \frac{4}{\pi^2}\sum_{n=0}^{\infty}\left[\frac{1}{(2n+1)^2}\exp\left(-(2n+1)^2\frac{\pi^2 D_{\mathrm{aq}}t}{4a^2}\right)\right] \tag{A2}$$

For $M = 1$, the mass in both streamtubes are the same (complete mixing). The short-term approximation, which is employed for the correction of α_{tv} in Eqs. (7)–(10) at high flow velocities, is:

$$M = \sqrt{\frac{D_{\mathrm{aq}}t}{a^2}\frac{4}{\pi}} \tag{A3}$$

Fig. A2 shows the short-term approximation fits for dimensionless times t' ($= D_{\mathrm{aq}}t/a^2$) of less than 0.4, where less than 70% of the mixing occurred (i.e. only 70% of the mass transferred after complete mixing is present in that streamtube containing initially no solute). This means that at this point, only a mass fraction of 0.7 is carried by this streamtube and able to participate in a dispersion process in the next pore throat, which results in a decrease of mechanical dispersion. For comparison, Taylor (1953) gives a dimensionless

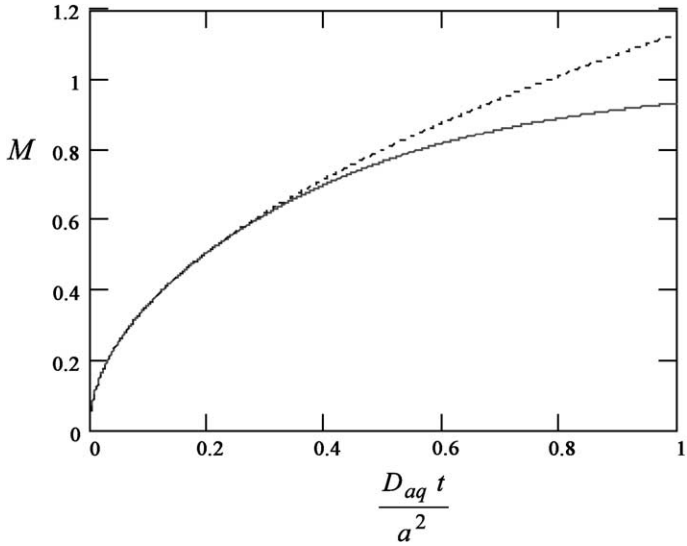

Fig. A2. Relative mass (M) diffused between the streamtubes vs. dimensionless time ($M=1$ denotes complete mixing). Comparison between Eq. (A2) (solid line) and the short-term approximation Eq. (A3) (dashed line).

time of 0.07 for a solute in a single capillary of circular shape to achieve a radial variation of the concentration of less than 1/e. Saffman (1959) provides an estimate for the complete mixing at $t'=1/8$ for diffusion in a pore based on the mean square displacement ($z^2=2D_{aq}t$; z denotes the radial diffusion distance) and the assumption of $z>1/2a$, which is similar to the approach presented in Eqs. (7)–(10) (with the difference that we assume $z=a$ equivalent to $t'=0.5$ for complete mixing). According to Fig. A1, mixing is not complete at $t'=1/8$ (at that time, Eq. (A3) yields $M=0.4$), but almost complete (80%) after a $t'=0.5$ (Fig. A2). The absence of complete mixing is also reported by Berkowitz et al. (1994), who did comprehensive numerical simulations of mass transfer at fracture intersections for Stokes flow and plug flow at different flow geometries. They found that mixing is incomplete already at local scale Peclet numbers ($v\ a/D_{aq}$) as low as 0.01 (corresponding to a dimensionless time of approximately 100 assuming $t=a/v$).

Eq. (A2) can be combined with the dispersivity measured at low flow velocities in order to account for the decreasing dispersivity with high increasing velocity, e.g.:

$$\alpha_{tv}=\frac{3}{16}dM \tag{A4}$$

Using the short-term approximation for M (Eq. (A3)), which is valid if mixing is less than 70%, and replacing the residence time in a pore by d/v_a yields:

$$\alpha_{tv}=\frac{3}{16}d\sqrt{\frac{D_{aq}d}{a^2v_a}\frac{4}{\pi}} \tag{A5}$$

which equals the corrected transverse dispersivity in Eq. (11) in the paper. Fig. A3 shows the effect of incomplete mixing with increasing flow velocity and compares Eqs. (A5) and

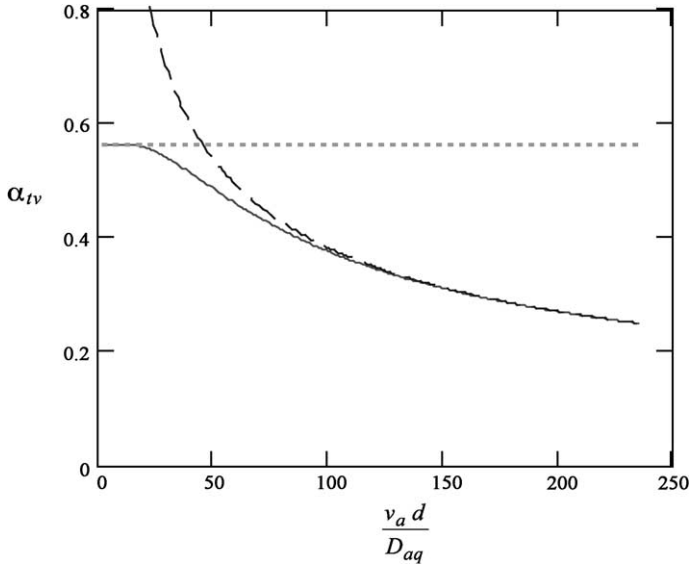

Fig. A3. α_{tv} [mm] vs. *Pe* number calculated from 3/16*d* (dotted line), accounting for incomplete mixing with Eq. (A3) (solid line), and with the short approximation (Eq. (A5); dashed line). Grain size $d = 1$ mm; $d/a = 6$ ($Pe^{crit} = 72$ for application of Eq. (5)).

(A4). The major unknown here is the ratio of *d* to *a,* which also appears if the critical Pe^{crit} number is defined above which incomplete mixing can be corrected by Eq. (5):

$$v_a^{crit} = \frac{2D_{aq}d}{a^2} \Rightarrow \frac{v_a^{crit}d}{2D_{aq}}\left(\frac{a}{d}\right)^2 \Rightarrow Pe^{crit} = 2\left(\frac{d}{a}\right)^2 \tag{A6}$$

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